

Final flag / password:

THE MAGIC OF WAND

For the plaintext submission field, use the full plaintext below only if they ask for the recovered message, not just the password/flag.

(i) RECOVERED PLAINTEXT

BREAKER OF THIS CODE WILL BE BLESSED BY THE SQUEAKY SPIRIT RESIDING IN THE HOLE
GO AHEAD AND FIND A WAY OF BREAKING THE SPELL ON HIM CAST BY THE EVIL KAFFAR THE
SPIRIT OF THE CAVE MAN IS ALWAYS WITH YOU FIND THE MAGIC WAND THAT WILL LET YOU
OUT OF THE CAVES IT WOULD MAKE YOU A MAGICIAN NO LESS THAN KAFFAR SPEAK THE
PASSWORD THE MAGIC OF WAND TO GO THROUGH

FINAL FLAG / PASSWORD

THE MAGIC OF WAND

(ii) ENCRYPTION ALGORITHM / CIPHER NAME

Two-layer classical cipher:

1. Monoalphabetic substitution cipher
2. Fixed block permutation / transposition cipher

(iii) ENCRYPTION KEYS

Permutation key:

43512

This is applied on blocks of 5 letters after substitution during decryption.

Substitution key used during decryption:

A TO T

B TO V

C TO I

D TO U

E TO C

F TO H

G TO G

H TO P

I TO Q

J TO B

K TO K

L TO S

M TO K

N TO R

P TO D

Q TO A

R TO W

S TO F

T TO L

U TO M

V TO E
W TO O
X TO Y
Y TO N

(iv) APPROACH FOR PROBLEM SOLUTION

First, the ciphertext was checked for randomness. The letter frequency looked similar to natural English, so it was likely a classical cipher rather than a modern encryption scheme.

Simple Caesar, affine, and direct substitution attempts did not produce readable text. Since the challenge mentioned two layers, the next assumption was that the ciphertext used substitution plus transposition.

The ciphertext letters were first converted using a monoalphabetic substitution mapping. This produced text that was still scrambled, but the scrambled groups looked close to English. Then the result was divided into blocks of 5 letters.

Each 5-letter block was rearranged using the permutation key 43512. This means the output block is formed by taking letters in the order 4, 3, 5, 1, 2 from each substituted block.

After applying the substitution and the 5-letter permutation, the plaintext became readable. The recovered message revealed the password:

THE MAGIC OF WAND

Note: the given ciphertext appears to have a small typo at the very end. The supplied ending decrypts to THRHUOG, but the intended plaintext is THROUGH.